

Effects of Sibship Size on Children's Foundational Learning Skills

Table 5.1 The Effect of Sibship Size on Children's Numeracy and Literacy Skills

	(1)	(2)	(3)	(4)
VARIABLES	Lit OLS	Lit 2SLS	Num OLS	Num 2SLS
Panel A				
Number of children (5-17)	-0.019*** (0.005)	-0.081** (0.041)	-0.046*** (0.005)	-0.121** (0.063)
First stage (Twins)		0.511*** (0.059)		0.511*** (0.059)
F statistic		40		40
R-squared	0.429	0.341	0.392	0.258
Panel B				
Parents went to school	0.036** (0.028)		0.041 (0.031)	
Someone read books	0.054* (0.031)		0.061** (0.033)	
Attended PTA	0.019 (0.025)		0.017 (0.028)	
Homework help	-0.051 (0.031)		-0.071 (0.032)	
R-squared	0.408		0.310	
Panel C				
VARIABLES	Parents went to school	Someone read books	Attended PTA	Homework help
Number of children (5-17)	-0.019** (0.007)	-0.010 (0.008)	-0.017* (0.011)	-0.005 (0.009)
R-squared	0.261	0.275	0.272	0.301
Observation	2385	2385	2385	2385
Controls	YES	YES	YES	YES
Locality (Cluster Fixed effects)	YES	YES	YES	YES

Robust standard errors in parentheses clustered at 420 clusters *** p<0.01, ** p<0.05, * p<0.1

Figure 5.1

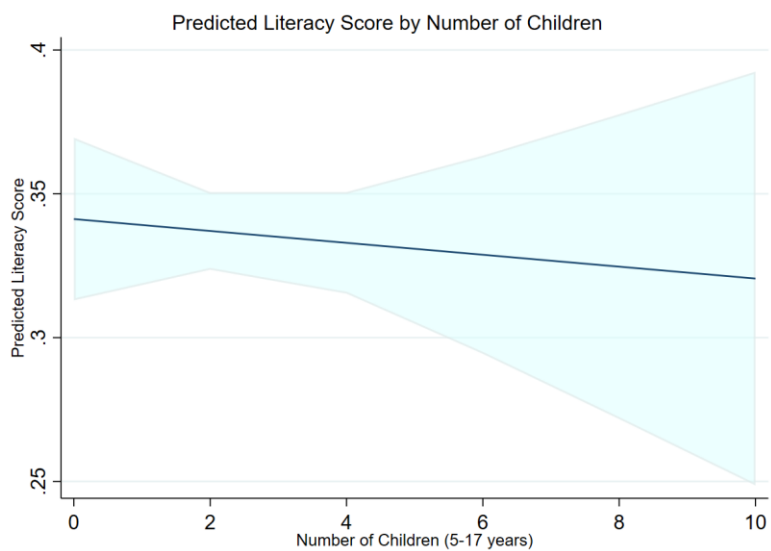


Figure 5.2

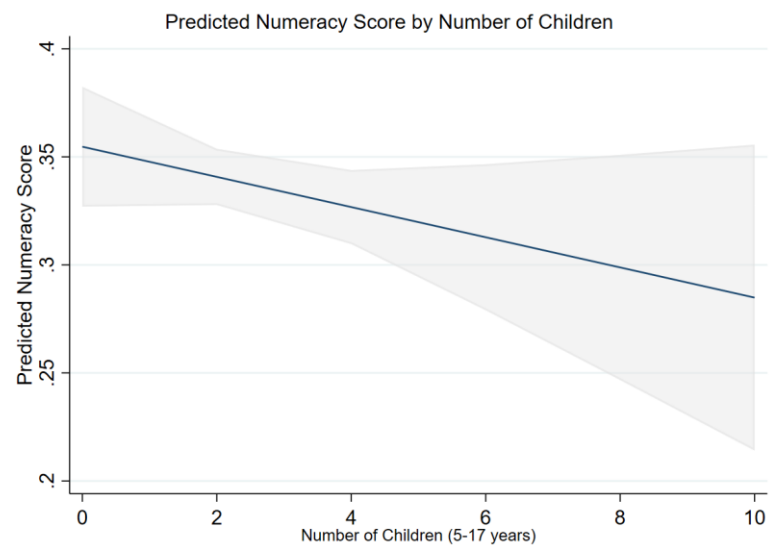


Table 5.2 Robustness test the effect of family size on children’s reading and mathematics test scores using a different data set (LaNA 2023 data)

VARIABLES	(1) Log (Reading Score)	(2) Log (Maths Score)
Family size	-0.048*** (0.016)	-0.026* (0.013)
Constant	3.238*** (0.091)	3.339*** (0.078)
Observations	2,526	2,526
R-squared	0.232	0.242
Child Characteristics	YES	YES
Family Characteristics	YES	YES
School Fixed effects	YES	YES

Robust standard errors in parentheses clustered at the school level *** p<0.01, ** p<0.05, * p<0.1

Heterogeneous Effects of Sibship Size on Children's Foundational Learning Skills

Table 5.3 The heterogeneous effects of Sibship size on Literacy skills by gender, area of residence, and wealth

Dep. Var. Literacy skills	Gender		Locality Type		By Wealth	
	Female	Male	urban	rural/camp	below mean	above mean
Panel A (LPM)						
Number of children (5 -17)	-0.012* (0.007)	-0.028*** (0.008)	-0.011* (0.006)	0.005 (0.008)	-0.018** (0.007)	-0.012 (0.010)
Constant	-2.198*** (0.091)	-1.981*** (0.079)	-2.141*** (0.067)	-2.135*** (0.086)	-1.961*** (0.088)	-2.254*** (0.070)
Panel B (2SLS)						
Number of children (5 -17)	-0.035* (0.020)	-0.080*** (0.026)	-0.045** (0.020)	0.012 (0.031)	-0.055** (0.024)	-0.040 (0.036)
Observations	1169	1216	1445	940	1160	1225
R-squared	0.522	0.478	0.448	0.462	0.470	0.501
Controls	YES	YES	YES	YES	YES	YES
Locality (Cluster Fixed effects)	YES	YES	YES	YES	YES	YES

Robust standard errors in parentheses clustered at 420 clusters *** p<0.01, ** p<0.05, * p<0.1

Table 5.4 The heterogeneous effects of Sibship size on numeracy skills by gender, area of residence, and wealth

Dep. Var. Numeracy skills	Gender		Locality Type		By Wealth	
	Female	Male	urban	rural/camp	below mean	above mean
Panel A (LPM)						
Number of children (5 -17)	-0.009 (0.007)	-0.028*** (0.009)	-0.031* (0.017)	0.005 (0.008)	-0.021*** (0.007)	-0.010 (0.007)
Constant	-2.057*** (0.097)	-1.862*** (0.080)	-1.929*** (0.070)	-2.080*** (0.089)	-1.767*** (0.092)	-2.095*** (0.074)
Panel B (2SLS)						
Number of children (5 -17)	-0.044 (0.022)	-0.066** (0.032)	-0.055** (0.027)	0.016 (0.109)	-0.039** (0.019)	-0.025 (0.072)
Observations	1169	1216	1445	940	1160	1225
R-squared	0.464	0.471	0.402	0.418	0.404	0.467
Controls	YES	YES	YES	YES	YES	YES
Locality (Cluster Fixed effects)	YES	YES	YES	YES	YES	YES

Robust standard errors in parentheses clustered at 420 clusters *** p<0.01, ** p<0.05, * p<0.1

Table 5.5 The heterogeneous effects of Sibship size on numeracy skills and literacy skills by birth order

Vars	Numeracy skills		Literacy skills	
	1st/2nd	other	1st/2nd	other
Panel A (LPM)				
Number of children (5 -17)	-0.052*** (0.009)	-0.030*** (0.008)	-0.024** (0.011)	-0.011*** (0.003)
Panel B (2SLS)				
Number of children (5 -17)	-0.133* (0.075)	-0.089** (0.039)	-0.068* (0.041)	-0.081** (0.041)
F statistic	21	17	21	17
Observations	1326	1059	1326	1059
R-squared	0.201	0.198	0.317	0.212
Controls	YES	YES	YES	YES
Locality (Cluster Fixed effects)	YES	YES	YES	YES

Robust standard errors in parentheses clustered at 420 clusters *** p<0.01, ** p<0.05, * p<0.1

The Influence of Sibship Size on Children's Mental Health

Table 5.6 The effect of sibship size on child mental health

VARIABLES	(1) Difficulty controlling behavior	(2) Difficulty concentrating	(3) Difficulty remembering	(4) Nervous/worried (How often)	(5) Sad/Depressed (How often)
Panel A					
Number of children (5-17)	0.005 (0.007)	0.002 (0.006)	0.012** (0.006)	-0.049** (0.025)	-0.026 (0.021)
Panel B (2SLS)					
2SLS (twins)	0.010 (0.014)	0.014* (0.008)	0.027** (0.013)	-0.083** (0.038)	-0.039 (0.026)
Observations	2,385	2,385	2,385	2,385	2,385
R-squared	0.259	0.231	0.224	0.464	0.297
Controls	YES	YES	YES	YES	YES
Locality (Cluster Fixed effects)	YES	YES	YES	YES	YES

Robust standard errors in parentheses clustered at 420 clusters *** p<0.01, ** p<0.05, * p<0.1

Table 5.7 Mechanism

VARIABLES	Level 1					level 2
	Difficulty controlling behavior	Difficulty concentrating	Difficulty remembering	Nervous/worried (How often)	Sad/Depressed (How often)	violent discipline
violent discipline	0.012*	0.006	0.010*	0.027	-0.023	
	(0.007)	(0.005)	(0.005)	(0.021)	(0.018)	
Number of children (5-17)						0.067***
						(0.021)
Overall Family Size						0.065***
						(0.015)
Number of Children less than 5						0.016
						(0.034)
Constant	0.143	-0.059	-0.086	-0.997	0.062	2.150***
	(0.247)	(0.149)	(0.146)	(0.654)	(0.575)	(0.742)
Observations	2,385	2,385	2,385	2,385	2,385	2,385
R-squared	0.109	0.039	0.187	0.374	0.157	0.166
Controls	YES	YES	YES	YES	YES	YES
Locality (Cluster Fixed effects)	YES	YES	YES	YES	YES	YES

Robust standard errors in parentheses clustered at 420 clusters *** p<0.01, ** p<0.05, * p<0.1

Figure 5.3

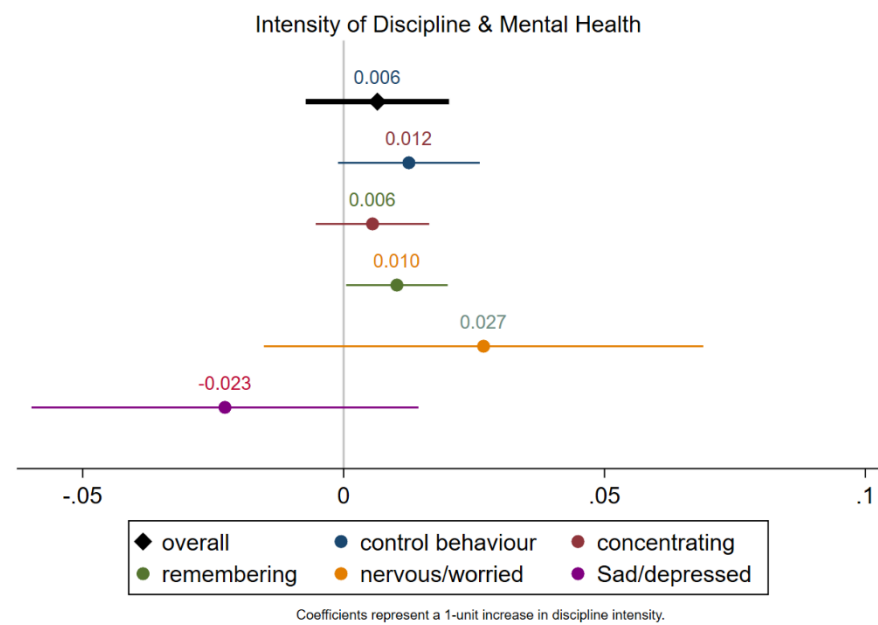
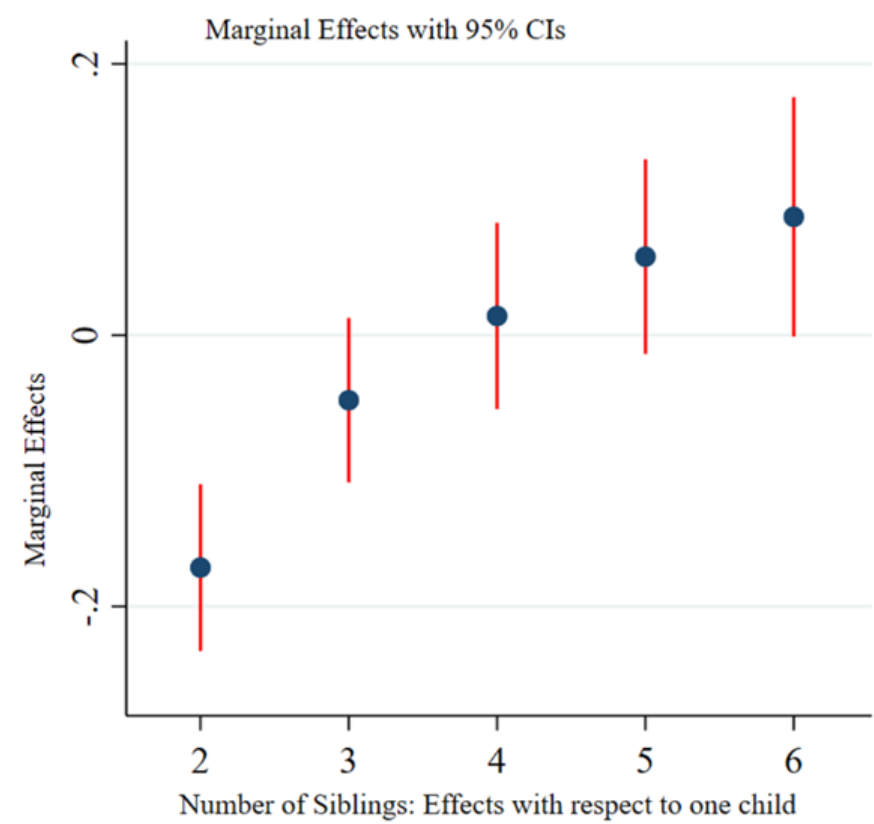


Figure 5.4



Socio-Economic Differences in the Effects of Sibship Size on Children's Mental Health

Table 5.8 by wealth

	(1)	(2)	(3)	(4)	(5)
	Difficulty controlling behavior	Difficulty concentrating	Difficulty remembering	Nervous/worried (How often)	Sad/Depressed (How often)
VARIABLES					
Panel A (below wealth mean)					
Number of children (5-17)	0.003 (0.010)	0.004 (0.008)	0.008 (0.007)	-0.090*** (0.034)	-0.033 (0.029)
Overall Family Size	-0.002 (0.005)	-0.007** (0.003)	-0.010** (0.005)	-0.004 (0.018)	-0.037*** (0.014)
Number of Children less than 5	0.012 (0.017)	-0.003 (0.011)	0.006 (0.010)	-0.022 (0.045)	0.068** (0.033)
Observations	876	876	876	876	876
R-squared	0.110	0.086	0.124	0.390	0.200
Panel B (Above wealth mean)					
Number of children (5-17)	0.004 (0.009)	-0.004 (0.006)	0.014** (0.007)	-0.006 (0.024)	-0.012 (0.022)
Overall Family Size	0.005 (0.006)	0.002 (0.005)	-0.002 (0.004)	0.014 (0.016)	-0.032** (0.016)
Number of Children less than 5	0.007 (0.014)	0.011 (0.009)	-0.001 (0.009)	0.040 (0.039)	0.082** (0.034)
Observations	1,509	1,509	1,509	1,509	1,509
R-squared	0.119	0.048	0.068	0.373	0.122
Controls	YES	YES	YES	YES	YES
Locality (Cluster Fixed effects)	YES	YES	YES	YES	YES

Robust standard errors in parentheses clustered at 420 clusters *** p<0.01, ** p<0.05, * p<0.1

Figure 5.5

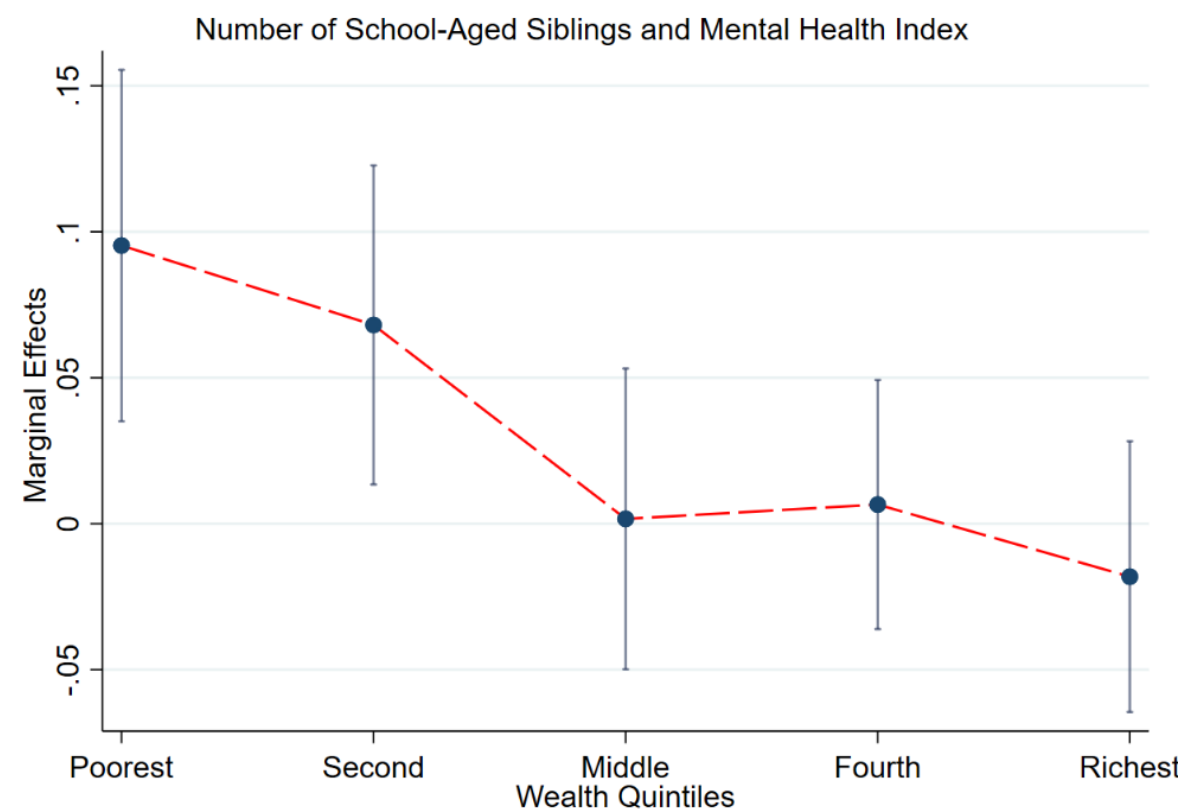


Table 5.9 The effect of Sibship Size on Child Mental health: Wealth Heterogeneity (Interaction)

VARIABLES	(1) Difficulty controlling behavior	(2) Difficulty concentrating	(3) Difficulty remembering	(4) Nervous/worried (How often)	(5) Sad/Depressed (How often)
Wealth Q2*Number of Children aged 5 - 17	0.008 (0.021)	-0.024 (0.014)	0.020 (0.014)	0.076 (0.061)	-0.062 (0.066)
Wealth Q3*Number of Children aged 5 - 17	0.028 (0.023)	0.014 (0.017)	0.011 (0.013)	0.146** (0.059)	-0.035 (0.068)
Wealth Q4*Number of Children aged 5 - 17	0.019 (0.020)	-0.015 (0.015)	0.031** (0.014)	0.087 (0.059)	-0.036 (0.063)
Wealth Q5*Number of Children aged 5 - 17	0.009 (0.019)	0.002 (0.015)	0.033*** (0.011)	0.160*** (0.060)	-0.016 (0.063)
Observations	2,385	2,385	2,385	2,385	2,385
R-squared	0.094	0.038	0.077	0.366	0.139
Controls	YES	YES	YES	YES	YES
Interaction vars	YES	YES	YES	YES	YES
Locality (Cluster Fixed effects)	YES	YES	YES	YES	YES

Robust standard errors in parentheses clustered at 420 clusters *** p<0.01, ** p<0.05, * p<0.1